# 10 月 21 日 輪読補足資料
# 多様体上の最適化：完備性と Manopt

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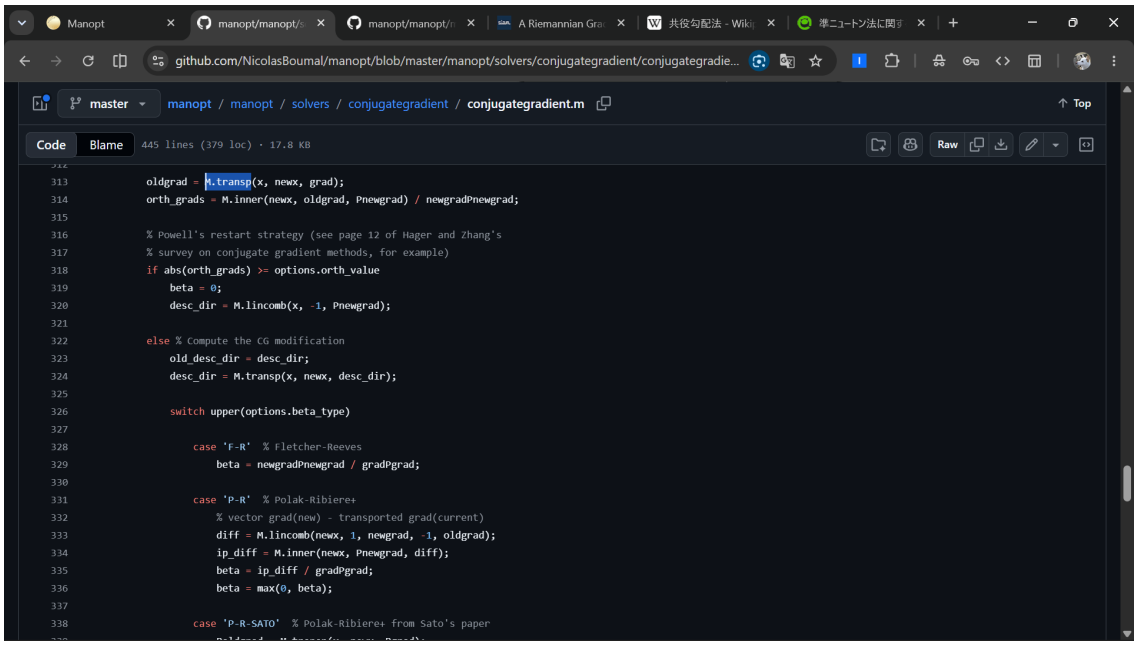
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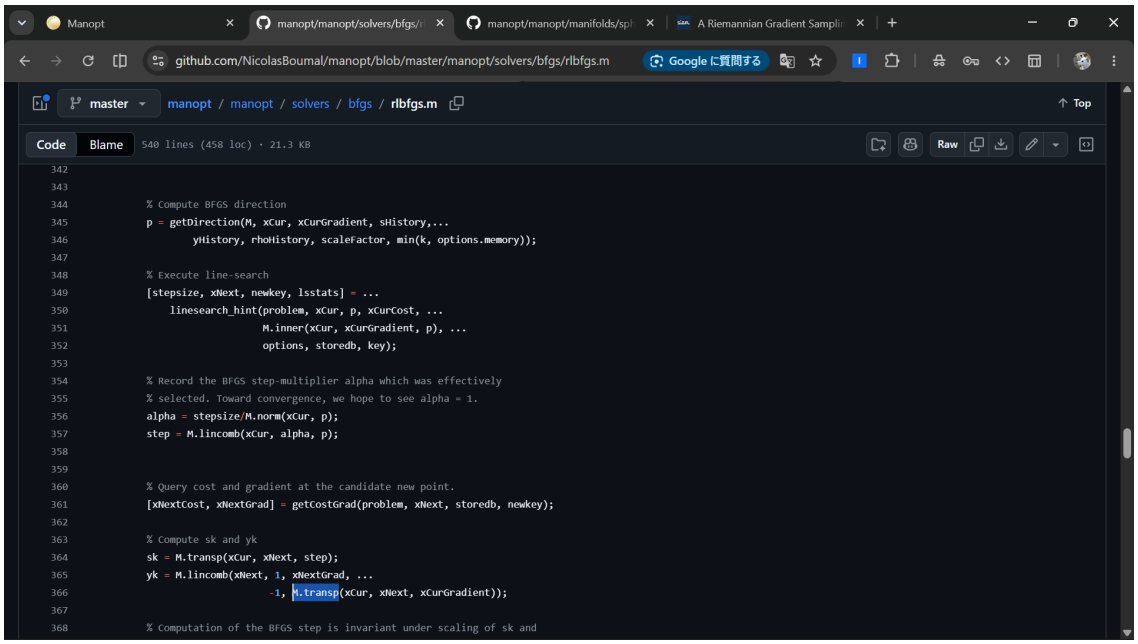
### 導入: なぜ transporter を考えたいのか?

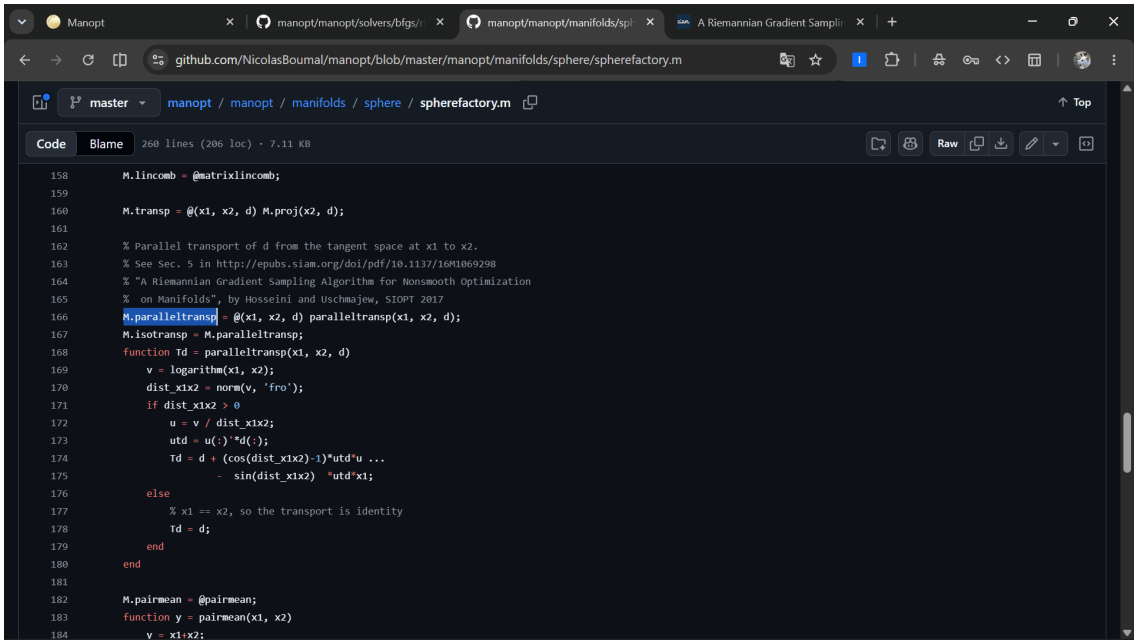
- 非線形共役勾配法 (nonlinear conjugate gradient, CG)
  - そもそもの共役勾配法は、対称正定値行列を係数とする連立一次方程式を解くためのアルゴリズム
  - (Solving $Ax = b$ for $A \succ 0$, i.e., minimizing $f(x) = \frac{1}{2}x^\top Ax - b^\top x$)
  - これは直感的にはおよそ次元分だけ共役方向 ($u^\top Av = 0$ という関係を満たすベクトルペア、つまり、$A$ というヘッセ行列で考えたときの直交関係。実際、$A = I$ なら、$u^\top v = 0$ となる) を取り除き続けて、解を求める
  - 非線形の場合も、局所的には二次関数に近いと考えて、CG の考え方で最適化し続ける。
  - 大事なこととして、この手法では各イテレーションで別の場所で勾配をとることになる (つまり、$u$ や $v$ というベクトルの始点が変わる)
  - “綺麗に” 現在の方向から、過去の共役方向の集積物を除去したい。
  - ユークリッド空間では自明
  - では、多様体上では?
- 準ニュートン法 (Broyden–Fletcher–Goldfarb–Shanno, BFGS)
  - ヘッセ行列を近似できれば大抵の場合に強い
  - 今持っているヘッセ行列の近似に、直近の 2 つの地点での勾配変化を反映させる操作が BFGS
  - $B_{k+1} = B_k - \frac{B_k s_k s_k^\top B_k}{s_k^\top B_k s_k} + \frac{y_k y_k^\top}{y_k^\top s_k}$, where $s_k = x_{k+1} - x_k$, $y_k = g_{k+1} - g_k$
  - これも、現在の勾配ベクトルが過去の勾配ベクトルからどう変化したか? が重要になる
  - “綺麗に” ベクトル同士の引き算がしたい
  - ユークリッド空間では自明

– では、多様体上では?

- ではこの “綺麗さ” をどう定義してどう担保しますか? というのが問題
- 結論としては、今回扱う parallel transport が理論的には最も綺麗で、内積を保存できる。実用的に、より軽く計算できる射影なども含めた一般化としての transporter を使うこともある。
- Manopt の実装 (リンク 1, リンク 2, リンク 3) でもそれが使われている。






- これまた大雑把に言えば、点ごとに接空間が変わることを考慮して、ベクトルに補正を掛けてから演算すればよい。
- その接ベクトル場の変化を測る仕組みが、今回は $\frac{D}{\mathrm{d}t}$ という共変微分の概念
- ただ、平行移動は経路を指定して初めて well-defined になる。地球上でベクトルを考える。赤道から直接北極に向かって運ぶのと、赤道から西に行ってから北極に向かうのでは、最終的なベクトルの向きが違う。
- 要は、経路も同時に定義で考慮する必要があるという意味
- その「経路」を、十分近い 2 点の間では、(大雑把に) 最短経路である一意な測地線に限定することで、始点と終点から平行移動が well-defined になる
- この経路の選択には、指数写像・対数写像が使える (教科書 p.272、命題 10.64)。
- では最短測地線の存在性の復習から始めましょう、というのが、今回の輪読の主旨

# 1 距離・完備性と Hopf–Rinow の定理

**前提の復習：距離と完備性**

$\mathcal{M}$ を連結な有限次元リーマン多様体とし、$\|v\|_x := \sqrt{\langle v, v\rangle_x}$ と書く。

- 区分的に滑らかな曲線 $c\colon [0,1] \to \mathcal{M}$ の長さと、リーマン距離を次で定義する。

$$L(c) := \int_0^1 \|c'(t)\|_{c(t)}\, \mathrm{d}t\,, \tag{S.1}$$

$$\operatorname{dist}(x,y) \coloneqq \inf_{\substack{c(0)=x\\ c(1)=y}} L(c). \tag{S.2}$$

- Cauchy 列とは、任意の $\varepsilon > 0$ に対し、十分大きい $m, n$ について $\operatorname{dist}(x_m, x_n) < \varepsilon$ となる点列である。すべての Cauchy 列が $\mathcal{M}$ 内に収束するとき、$\mathcal{M}$ は距離完備という。
- 曲線 $c$ に沿うベクトル場 $Z(t) \in \mathrm{T}_{c(t)}\mathcal{M}$ の共変微分 $\frac{D}{\mathrm{d}t}Z$ は、リーマン接続 (捻れがなく、計量と整合するアフィン接続)$\nabla$ から誘導される $\mathbb{R}$ 線形な演算である。スカラー関数 $h$ と多様体上のベクトル場 $X$ に対し、次の性質で特徴づけられる。

$$\frac{D}{\mathrm{d}t}(hZ) = h'Z + h\frac{D}{\mathrm{d}t}Z, \tag{S.3}$$

$$\frac{D}{\mathrm{d}t}(X \circ c)(t) = \nabla_{c'(t)}X. \tag{S.4}$$

  ユークリッド空間のリーマン部分多様体では、$\frac{D}{\mathrm{d}t}Z(t) = \mathrm{Proj}_{c(t)}(Z'(t))$、つまり通常の微分を接空間に射影したものである。
- 測地線とは、$\frac{D}{\mathrm{d}t}c'(t) = 0$ を満たす滑らかな曲線 $c$ である。すべての測地線が実数全体で定義された測地線に延長できるとき、$\mathcal{M}$ は測地的完備という。

### Theorem 10.8 (Hopf–Rinow)

**Theorem 10.8 (Hopf–Rinow).** A connected Riemannian manifold $\mathcal{M}$ is metrically complete if and only if it is geodesically complete. Additionally, $\mathcal{M}$ is complete (in either sense) if and only if its compact subsets are exactly its closed and bounded subsets.

### 定理の読み方

連結で境界のない有限次元リーマン多様体では、次の 3 条件が同値である。

(1) 閉かつ有界な部分集合はコンパクトである（HB）。
(2) 距離空間として完備である、すなわちすべての Cauchy 列が $\mathcal{M}$ 内に収束する（MC）。
(3) 任意の $p \in \mathcal{M}$ で指数写像 $\operatorname{Exp}_p$ が $\mathrm{T}_p\mathcal{M}$ 全体に定義される、すなわち測地的完備である（GC）。

さらに、いずれかが成立すれば、任意の 2 点 $x, y \in \mathcal{M}$ を結ぶ最短測地線 $c$ が存在し、距離の下限が $L(c) = \operatorname{dist}(x,y)$ として達成される。参考：Wikipedia「Hopf–Rinow theorem」

**2.4.2 Theorem (Hopf-Rinow).** *Let $(M, g)$ be a connected RMF, then the following conditions are equivalent:*

*(MC)* *The metric space $(M, d)$ is complete (i.e., every Cauchy sequence converges).*

*(GC')* *There is $p \in M$ such that $M$ is geodesically complete at $p$, i.e., $\exp_p$ is defined on all of $T_pM$.*

*(GC)* *$(M, g)$ is geodesically complete.*

*(HB)* *$M$ possesses the Heine-Borel property, i.e., every closed and bounded subset of $M$ is compact.*

**Hopf–Rinow の定理の一方向の証明：測地的完備なら距離完備**

$\mathcal{M}$ を連結で測地的完備（GC）なリーマン多様体とする。

**1.** GC $\Rightarrow$ GC′**：一点での測地的完備性。**点 $p \in \mathcal{M}$ を固定し、初期条件 $\gamma_v(0) = p$、$\gamma_v'(0) = v$ の測地線を $\gamma_v$ と書く。指数写像を $\mathrm{Exp}_p(v) \coloneqq \gamma_v(1)$ と定める。GC により、これは $\mathrm{T}_p\mathcal{M}$ 全体で定義される。このような点 $p$ が少なくとも一つ存在するという条件を GC′ と呼ぶ。

**2.** GC′ $\Rightarrow$ HB**：閉かつ有界な集合はコンパクト。GC′ から、$p$ と任意の点を結ぶ最短測地線が存在することは、既知とする（参考資料の補題 2.4.1）。**指数写像の連続性と、測地線の速さが一定であることも既知とする。$A \subseteq \mathcal{M}$ を閉かつ有界とし、すべての $q \in A$ に対して $\mathrm{dist}(p, q) \leq R$ となる $R > 0$ を取る。接空間の閉球を

$$K_R \coloneqq \{v \in \mathrm{T}_p\mathcal{M} : \|v\|_p \leq R\} \tag{S.5}$$

とおく。各 $q \in A$ への定速の最短測地線を $c_q \colon [0, 1] \to \mathcal{M}$ とすると、定速性と式 (S.1)、(S.2) から

$$\left\|c_q'(0)\right\|_p = L(c_q) = \mathrm{dist}(p, q) \leq R \tag{S.6}$$

なので、式 (S.5)、(S.6) と指数写像の定義から $A \subseteq \mathrm{Exp}_p(K_R)$ となる。**接空間**は有限次元の内積空間なので、Heine–Borel により $K_R$ はコンパクトである。その連続像 $\mathrm{Exp}_p(K_R)$ もコンパクトで、$A$ はその閉部分集合なのでコンパクトである（HB）。

**3.** HB $\Rightarrow$ MC**：Cauchy 列が収束する。**Cauchy 列 $(x_n)$ は有界なので、その点集合の閉包は HB によりコンパクトである。よって収束部分列を持ち、Cauchy 性と三角不等式から列全体も同じ極限に収束する。したがって $\mathcal{M}$ は距離完備（MC）である。

証明の参考：Kunzinger–Steinbauer、補題 2.4.1 と定理 2.4.2 の証明

# 2　指数写像と対数写像（10.2 節）

以下、教科書の主張を簡略表記する（青：今回の範囲、aqua：範囲外の前提事項）。

**前提：Definitions 3.42, 3.44（pp.37–38）**

**Definition 3.42.** The tangent bundle is

$$\mathrm{T}\mathcal{M} = \{(x, v) : x \in \mathcal{M} \text{ and } v \in \mathrm{T}_x\mathcal{M}\}. \tag{3.30}$$

**Definition 3.44.** A vector field is a map $V\colon \mathcal{M} \to \mathrm{T}\mathcal{M}$ with $V(x) \in \mathrm{T}_x\mathcal{M}$ for all $x$. Smooth vector fields form $\mathfrak{X}(\mathcal{M})$.

**前提：Definition 3.47（p.39）**

**Definition 3.47.** A retraction on a manifold $\mathcal{M}$ is a smooth map $\mathrm{R}\colon \mathrm{T}\mathcal{M} \to \mathcal{M}\colon (x, v) \mapsto \mathrm{R}_x(v)$ such that each curve $c(t) = \mathrm{R}_x(tv)$ satisfies $c(0) = x$ and $c'(0) = v$.

**前提：Definitions 5.36, 5.38, 5.42（pp.105–108）**

**Definition 5.36.** For a smooth curve $c\colon I \to \mathcal{M}$, velocity is $c' \in \mathfrak{X}(c)$ and intrinsic acceleration is $c'' = \frac{D}{\mathrm{d}t}c' \in \mathfrak{X}(c)$.

**Definition 5.38.** A geodesic on a Riemannian manifold is a smooth curve $c\colon I \to \mathcal{M}$ with $c''(t) = 0$ for all $t \in I$, where $I \subseteq \mathbb{R}$ is an open interval.

**Definition 5.42.** A retraction R is second-order if $c(t) = \mathrm{R}_x(tv)$ satisfies $c''(0) = 0$ for all $(x, v) \in \mathrm{T}\mathcal{M}$.

**Definition 10.16, Propositions 10.17–10.18（pp.264–265）**

For each $(x, v) \in \mathrm{T}\mathcal{M}$, let $\gamma_v$ be the unique maximal geodesic with

$$\gamma_v\colon I \to \mathcal{M}, \quad \text{with} \quad \gamma_v(0) = x \quad \text{and} \quad \gamma_v'(0) = v.$$

**Definition 10.16.** Define

$$\mathcal{O} = \{(x, v) \in \mathrm{T}\mathcal{M} : \gamma_v \text{ is defined on an interval containing } [0, 1]\}.$$

The exponential map $\mathrm{Exp}\colon \mathcal{O} \to \mathcal{M}$ is defined by

$$\mathrm{Exp}(x, v) = \mathrm{Exp}_x(v) = \gamma_v(1).$$

The restriction $\mathrm{Exp}_x$ is defined on $\mathcal{O}_x = \{v \in \mathrm{T}_x\mathcal{M} : (x, v) \in \mathcal{O}\}$.
Since $\gamma_{tv}(1) = \gamma_v(t)$ whenever either is defined,

$$\mathrm{Exp}_x(tv) = \gamma_v(t), \tag{10.3}$$

**Proposition 10.17.** The exponential map is smooth on its domain $\mathcal{O}$, which is open in $\mathrm{T}\mathcal{M}$.

**Proposition 10.18.** The exponential map is a second-order retraction, with a possibly restricted domain $\mathcal{O} \subseteq \mathrm{T}\mathcal{M}$.

### Definitions 10.19–10.20 (pp.265–266)

**Definition 10.19.** The injectivity radius of a Riemannian manifold $\mathcal{M}$ at a point $x$, denoted by $\mathrm{inj}(x)$, is the supremum over radii $r > 0$ such that $\mathrm{Exp}_x$ is defined and is a diffeomorphism on the open ball $B(x, r) = \{v \in \mathrm{T}_x\mathcal{M} : \|v\|_x < r\}$. By the inverse function theorem, $\mathrm{inj}(x) > 0$.

**Definition 10.20.** For $x \in \mathcal{M}$, let $\mathrm{Log}_x$ denote the logarithmic map at $x$,

$$\mathrm{Log}_x(y) = \operatorname*{arg\,min}_{v \in \mathcal{O}_x} \|v\|_x \quad \text{subject to} \quad \mathrm{Exp}_x(v) = y, \tag{10.5}$$

with domain such that this is uniquely defined.

### Proposition 10.22 (p.267)

**Proposition 10.22.** If $\|v\|_x < \mathrm{inj}(x)$, the geodesic $c(t) = \mathrm{Exp}_x(tv)$ on the interval $[0, 1]$ is the minimizing curve connecting $x$ to $y = \mathrm{Exp}_x(v)$, unique up to parameterization. In particular, $\mathrm{dist}(x, y) = \|v\|_x$, and $\mathrm{Log}_x(y) = v$.

### Proposition 10.23 (p.267)

**Proposition 10.23.** Let $\mathcal{M}$ be a manifold with retraction R defined on a neighborhood $\mathcal{O}$ of the zero section of $\mathrm{T}\mathcal{M}$. Consider the following map:

$$E \colon \mathcal{O} \to \mathcal{M} \times \mathcal{M} \colon (x, v) \mapsto E(x, v) = (x, \mathrm{R}_x(v)). \tag{10.7}$$

If $\mathcal{T} \subseteq \mathcal{O}$ is open in $\mathrm{T}\mathcal{M}$ such that, for all $x$, $\mathrm{R}_x$ is a diffeomorphism on $\mathcal{T}_x = \{v \in \mathrm{T}_x\mathcal{M} : (x, v) \in \mathcal{T}\}$, then $\mathcal{V} = E(\mathcal{T})$ is open in $\mathcal{M} \times \mathcal{M}$ and $E \colon \mathcal{T} \to \mathcal{V}$ is a diffeomorphism.

### Proposition 10.24, Corollary 10.25 (p.268)

**Proposition 10.24.** On a Riemannian manifold $\mathcal{M}$, the injectivity radius function $\mathrm{inj} \colon \mathcal{M} \to (0, \infty]$ is continuous.

**Corollary 10.25.** The map $(x, v) \mapsto (x, \mathrm{Exp}_x(v))$ is a diffeomorphism from

$$\mathcal{T} = \{(x, v) \in \mathrm{T}\mathcal{M} : \|v\|_x < \mathrm{inj}(x)\}$$

to

$$\mathcal{V} = \{(x, y) \in \mathcal{M} \times \mathcal{M} : \mathrm{dist}(x, y) < \mathrm{inj}(x)\}.$$

Its inverse is $(x, y) \mapsto (x, \mathrm{Log}_x(y))$, smooth from $\mathcal{V}$ to $\mathcal{T}$.

**Corollary 10.27（pp.268–269）**

For continuous $\Delta\colon \mathcal{M} \to (0,\infty]$, set

$$\mathcal{T} = \{(x,v) \in \mathrm{T}\mathcal{M} : \|v\|_x < \Delta(x)\}. \tag{10.8}$$

**Corollary 10.27.** For any retraction R, the map $(x,v) \mapsto (x, \mathrm{R}_x(v))$ is a diffeomorphism onto its image on some zero-section neighborhood $\mathcal{T}$ of the form (10.8), with respect to any chosen Riemannian metric.

## 3　平行移動（10.3 節）

**前提：Definition 5.20（p.97）**

**Definition 5.20.** A connection is an operator $\nabla\colon \mathfrak{X}(\mathcal{M}) \times \mathfrak{X}(\mathcal{M}) \to \mathfrak{X}(\mathcal{M})\colon (U,V) \mapsto \nabla_U V$ satisfying, for smooth fields $U,V,W$, smooth functions $f,g \in \mathfrak{F}(\mathcal{M})$ and $a,b \in \mathbb{R}$:

1. $\mathfrak{F}(\mathcal{M})$-linearity in $U$: $\nabla_{fU+gW}V = f\nabla_U V + g\nabla_W V$;
2. $\mathbb{R}$-linearity in $V$: $\nabla_U(aV+bW) = a\nabla_U V + b\nabla_U W$; and
3. Leibniz rule: $\nabla_U(fV) = (Uf)V + f\nabla_U V$.

Here $\nabla_U V$ is the covariant derivative of $V$ along $U$.

**前提：Definition 5.5, Theorem 5.6（p.89）**

**Definition 5.5.** For smooth fields $U,V$ and a smooth function $f$ on an open $\mathcal{U} \subseteq \mathcal{M}$, define:

- $Uf \in \mathfrak{F}(\mathcal{U})$ such that $(Uf)(x) = \mathrm{D}f(x)[U(x)]$;
- $[U,V]\colon \mathfrak{F}(\mathcal{U}) \to \mathfrak{F}(\mathcal{U})$ such that $[U,V]f = U(Vf) - V(Uf)$; and
- $\langle U,V\rangle \in \mathfrak{F}(\mathcal{M})$ such that $\langle U,V\rangle(x) = \langle U(x), V(x)\rangle_x$.

**Theorem 5.6.** On a Riemannian manifold, the Levi-Civita connection is the unique connection satisfying, for all smooth fields $U,V,W$:

4. Symmetry: $[U,V]f = (\nabla_U V - \nabla_V U)f$ for all $f \in \mathfrak{F}(\mathcal{M})$; and
5. Compatibility with the metric: $U\langle V,W\rangle = \langle \nabla_U V, W\rangle + \langle V, \nabla_U W\rangle$.

**前提：Definition 5.28, Theorem 5.29（p.101）**

**Definition 5.28.** For a smooth curve $c\colon I \to \mathcal{M}$ on an open interval, $\mathfrak{X}(c)$ denotes

smooth maps $Z\colon I \to \mathrm{T}\mathcal{M}$ with $Z(t) \in \mathrm{T}_{c(t)}\mathcal{M}$.

**Theorem 5.29.** A connection $\nabla$ uniquely induces $\frac{D}{\mathrm{d}t}\colon \mathfrak{X}(c) \to \mathfrak{X}(c)$ satisfying, for $Y, Z \in \mathfrak{X}(c)$, $U \in \mathfrak{X}(\mathcal{M})$, smooth $g$ and $a, b \in \mathbb{R}$:

1. $\mathbb{R}$-linearity: $\frac{D}{\mathrm{d}t}(aY + bZ) = a\frac{D}{\mathrm{d}t}Y + b\frac{D}{\mathrm{d}t}Z$;
2. Leibniz rule: $\frac{D}{\mathrm{d}t}(gZ) = g'Z + g\frac{D}{\mathrm{d}t}Z$;
3. Chain rule: $\frac{D}{\mathrm{d}t}(U \circ c)(t) = \nabla_{c'(t)}U$ for all $t \in I$.

If $\nabla$ is metric-compatible, also:

4. Product rule: $\frac{\mathrm{d}}{\mathrm{d}t}\langle Y, Z\rangle = \langle \frac{D}{\mathrm{d}t}Y, Z\rangle + \langle Y, \frac{D}{\mathrm{d}t}Z\rangle$,

where $\langle Y, Z\rangle \in \mathfrak{F}(I)$ is defined by $\langle Y, Z\rangle(t) = \langle Y(t), Z(t)\rangle_{c(t)}$.

## Definition 10.33, Theorem 10.34, Definition 10.35 (p.271)

**Definition 10.33.** A vector field $Z \in \mathfrak{X}(c)$ such that $\frac{D}{\mathrm{d}t}Z = 0$ is parallel.

**Theorem 10.34.** Given a connection, a smooth curve $c\colon I \to \mathcal{M}$, $t_0 \in I$ and $u \in \mathrm{T}_{c(t_0)}\mathcal{M}$, there is a unique parallel $Z \in \mathfrak{X}(c)$ with $Z(t_0) = u$.

**Definition 10.35.** Parallel transport along $c$ is

$$\mathrm{PT}^c_{t_1 \leftarrow t_0}\colon \mathrm{T}_{c(t_0)}\mathcal{M} \to \mathrm{T}_{c(t_1)}\mathcal{M}$$

defined by $\mathrm{PT}^c_{t_1 \leftarrow t_0}(u) = Z(t_1)$ for the parallel field $Z$ with $Z(t_0) = u$.

Parallel transport from $x$ to $y$ depends on the choice of curve connecting $x$ and $y$.

## Proposition 10.36 (p.272)

**Proposition 10.36.** The parallel transport operator $\mathrm{PT}^c_{t_1 \leftarrow t_0}$ is linear. It satisfies $\mathrm{PT}^c_{t_2 \leftarrow t_1} \circ \mathrm{PT}^c_{t_1 \leftarrow t_0} = \mathrm{PT}^c_{t_2 \leftarrow t_0}$, $\mathrm{PT}^c_{t \leftarrow t} = \mathrm{id}$, and $(\mathrm{PT}^c_{t_1 \leftarrow t_0})^{-1} = \mathrm{PT}^c_{t_0 \leftarrow t_1}$. If $\nabla$ is metric-compatible, parallel transport is an isometry:

$$\forall u, v \in \mathrm{T}_{c(t_0)}\mathcal{M}, \quad \langle u, v\rangle_{c(t_0)} = \left\langle \mathrm{PT}^c_{t_1 \leftarrow t_0}(u), \mathrm{PT}^c_{t_1 \leftarrow t_0}(v)\right\rangle_{c(t_1)}.$$

Its adjoint then equals its inverse.

## Proposition 10.37 (p.273)

**Proposition 10.37.** For a smooth curve $c\colon I \to \mathcal{M}$, $Z \in \mathfrak{X}(c)$ and $t_0 \in I$, set $z(t) = \mathrm{PT}^c_{t_0 \leftarrow t}Z(t) \in \mathrm{T}_{c(t_0)}\mathcal{M}$. Then,

$$\frac{D}{\mathrm{d}t}Z(t) = \mathrm{PT}^c_{t \leftarrow t_0}\left(\frac{\mathrm{d}}{\mathrm{d}t}z(t)\right) = \lim_{\delta \to 0}\frac{\mathrm{PT}^c_{t \leftarrow t+\delta}Z(t+\delta) - Z(t)}{\delta}.$$

## 4　Lipschitz 条件と Taylor 展開（10.4 節）

**前提：Definition 3.58, Definition 5.14, Proposition 5.15（pp.42, 95）**

**Definition 3.58.** For smooth $f\colon \mathcal{M} \to \mathbb{R}$ on a Riemannian manifold, $\operatorname{grad} f$ is uniquely defined by

$$\forall (x, v) \in \mathrm{T}\mathcal{M}, \qquad \mathrm{D}f(x)[v] = \langle v, \operatorname{grad} f(x)\rangle_x, \tag{3.34}$$

**Definition 5.14.** With the Riemannian connection $\nabla$, the Hessian of smooth $f$ is $\operatorname{Hess} f(x)\colon \mathrm{T}_x\mathcal{M} \to \mathrm{T}_x\mathcal{M}$, $\operatorname{Hess} f(x)[u] = \nabla_u \operatorname{grad} f$. On vector fields, $\operatorname{Hess} f[U] = \nabla_U \operatorname{grad} f$.

**Proposition 5.15.** The Hessian is self-adjoint: $\langle \operatorname{Hess} f(x)[u], v\rangle_x = \langle u, \operatorname{Hess} f(x)[v]\rangle_x$ for all $x$ and $u, v \in \mathrm{T}_x\mathcal{M}$.

**10.4 節の規約、Definition 10.40, Propositions 10.41, 10.43（pp.274–276）**

Throughout Section 10.4, manifolds are Riemannian, metric subscripts are omitted, and differentiability assumptions are stated explicitly.

For metric spaces $A, B$ and $L \geq 0$, a map $F\colon A \to B$ is $L$-Lipschitz if

$$\forall x, y \in A, \qquad \operatorname{dist}_B(F(x), F(y)) \leq L \operatorname{dist}_A(x, y), \tag{10.12}$$

**Definition 10.40.** A function $f\colon \mathcal{M} \to \mathbb{R}$ on a connected manifold $\mathcal{M}$ is $L$-Lipschitz continuous if

$$\forall x, y \in \mathcal{M}, \qquad |f(x) - f(y)| \leq L \operatorname{dist}(x, y), \tag{10.13}$$

If $\mathcal{M}$ is disconnected, we require the condition to hold on each connected component separately.

**Proposition 10.41.** A function $f\colon \mathcal{M} \to \mathbb{R}$ is $L$-Lipschitz iff

$$\forall (x, s) \in \mathcal{O}, \qquad |f(\operatorname{Exp}_x(s)) - f(x)| \leq L\|s\|, \tag{10.14}$$

Here $\mathcal{O} = \operatorname{dom}(\operatorname{Exp})$.

**Proposition 10.43.** If $f$ has a continuous gradient, it is $L$-Lipschitz iff

$$\forall x \in \mathcal{M}, \qquad \|\operatorname{grad} f(x)\| \leq L. \tag{10.15}$$

**Definition 10.44, Proposition 10.45（p.277）**

**Definition 10.44.** A vector field $V$ on a connected manifold $\mathcal{M}$ is $L$-Lipschitz

continuous if, for all $x, y \in \mathcal{M}$ with $\operatorname{dist}(x, y) < \operatorname{inj}(x)$,

$$\|\mathrm{PT}_{0\leftarrow 1}^{\gamma} V(y) - V(x)\| \leq L \operatorname{dist}(x, y), \tag{10.16}$$

where $\gamma\colon [0, 1] \to \mathcal{M}$ is the unique minimizing geodesic connecting $x$ to $y$. If $\mathcal{M}$ is disconnected, we require the condition on each connected component.

**Proposition 10.45.** A vector field $V$ is $L$-Lipschitz iff

$$\forall (x, s) \in \mathcal{O}, \qquad \left\| P_s^{-1} V(\operatorname{Exp}_x(s)) - V(x) \right\| \leq L\|s\|, \tag{10.17}$$

where $\mathcal{O} \subseteq \mathrm{T}\mathcal{M}$ is the domain of Exp and $P_s$ denotes parallel transport along $\gamma(t) = \operatorname{Exp}_x(ts)$ from $t = 0$ to $t = 1$.

### Proposition 10.46, Corollary 10.47 (pp.278–279)

**Proposition 10.46.** A continuously differentiable vector field $V$ is $L$-Lipschitz iff

$$\forall (x, s) \in \mathrm{T}\mathcal{M}, \qquad \|\nabla_s V\| \leq L\|s\|, \tag{10.18}$$

where $\nabla$ is the Riemannian connection. Then, for any smooth curve $c\colon [0, 1] \to \mathcal{M}$ from $x$ to $y$,

$$\|\mathrm{PT}_{0\leftarrow 1}^{c} V(y) - V(x)\| \leq L \cdot L(c). \tag{10.19}$$

**Corollary 10.47.** For twice continuously differentiable $f$, $\operatorname{grad} f$ is $L$-Lipschitz iff, for all $x$,

$$\|\operatorname{Hess} f(x)\| = \max_{\substack{s \in \mathrm{T}_x\mathcal{M} \\ \|s\|=1}} \|\operatorname{Hess} f(x)[s]\| \leq L.$$

### Definition 10.49, Proposition 10.50 (pp.279–280)

**Definition 10.49.** For each $x \in \mathcal{M}$, let $H(x)\colon \mathrm{T}_x\mathcal{M} \to \mathrm{T}_x\mathcal{M}$ be linear. If $\mathcal{M}$ is connected, we say $H$ is $L$-Lipschitz continuous if for all $x, y \in \mathcal{M}$ such that $\operatorname{dist}(x, y) < \operatorname{inj}(x)$ we have

$$\|\mathrm{PT}_{0\leftarrow 1}^{\gamma} \circ H(y) \circ \mathrm{PT}_{1\leftarrow 0}^{\gamma} - H(x)\| \leq L \operatorname{dist}(x, y), \tag{10.20}$$

where $\|\cdot\|$ denotes the operator norm with respect to the Riemannian metric, and $\gamma\colon [0, 1] \to \mathcal{M}$ is the unique minimizing geodesic connecting $x$ to $y$. If $\mathcal{M}$ is disconnected, we require the condition on each connected component.

**Proposition 10.50.** For each $x \in \mathcal{M}$, let $H(x)\colon \mathrm{T}_x\mathcal{M} \to \mathrm{T}_x\mathcal{M}$ be linear. The map $H$ is $L$-Lipschitz iff

$$\forall (x, s) \in \mathcal{O}, \qquad \left\| P_s^{-1} \circ H(\operatorname{Exp}_x(s)) \circ P_s - H(x) \right\| \leq L\|s\|, \tag{10.21}$$

where $\mathcal{O} \subseteq \mathrm{T}\mathcal{M}$ is the domain of Exp and $P_s$ denotes parallel transport along $\gamma(t) = \mathrm{Exp}_x(ts)$ from $t = 0$ to $t = 1$.

**Proposition 10.53（pp.280–281）**

For $(x, s) \in \mathcal{O}$ and $\gamma(t) = \mathrm{Exp}_x(ts)$, write

$$P_{ts} = \mathrm{PT}^{\gamma}_{t \leftarrow 0} \tag{10.24}$$

for parallel transport along $\gamma$.

**Proposition 10.53.** Let $f \colon \mathcal{M} \to \mathbb{R}$ be continuously differentiable. Let $\gamma(t) = \mathrm{Exp}_x(ts)$ be defined on $[0, 1]$ and assume there exists $L \geq 0$ such that, for all $t \in [0, 1]$,

$$\left\| P_{ts}^{-1} \operatorname{grad} f(\gamma(t)) - \operatorname{grad} f(x) \right\| \leq L\|ts\|.$$

Then,

$$|f(\mathrm{Exp}_x(s)) - f(x) - \langle s, \operatorname{grad} f(x) \rangle| \leq \frac{L}{2}\|s\|^2.$$

**Proposition 10.55（p.282）**

**Proposition 10.55.** Let $f \colon \mathcal{M} \to \mathbb{R}$ be twice continuously differentiable. Let $\gamma(t) = \mathrm{Exp}_x(ts)$ be defined on $[0, 1]$ and assume there exists $L \geq 0$ such that, for all $t \in [0, 1]$,

$$\left\| P_{ts}^{-1} \circ \operatorname{Hess} f(\gamma(t)) \circ P_{ts} - \operatorname{Hess} f(x) \right\| \leq L\|ts\|.$$

Then,

$$\left| f(\mathrm{Exp}_x(s)) - f(x) - \langle s, \operatorname{grad} f(x) \rangle - \frac{1}{2}\langle s, \operatorname{Hess} f(x)[s] \rangle \right| \leq \frac{L}{6}\|s\|^3,$$

$$\left\| P_s^{-1} \operatorname{grad} f(\mathrm{Exp}_x(s)) - \operatorname{grad} f(x) - \operatorname{Hess} f(x)[s] \right\| \leq \frac{L}{2}\|s\|^2.$$

## 5 Transporters（10.5 節）

**前提：Definition 3.68, Proposition 8.51（pp.46, 200）**

**Definition 3.68.** A local frame near $x$ consists of smooth fields $W_1, \ldots, W_n$ such that $W_1(y), \ldots, W_n(y)$ form a basis of $\mathrm{T}_y\mathcal{M}$ at every nearby $y$, where $n = \dim \mathcal{M}$. Let $(\mathcal{U}, \varphi)$ be a chart on a $d$-dimensional manifold $\mathcal{M}$. Its coordinate vector fields are

$$W_i(x) = \left[ t \mapsto \varphi^{-1}(\varphi(x) + te_i) \right], \qquad i = 1, \ldots, d, \tag{8.15}$$

where $e_1, \ldots, e_d$ are the standard basis of $\mathbb{R}^d$.

**Proposition 8.51.** The coordinate fields $W_1, \ldots, W_d \in \mathfrak{X}(\mathcal{U})$ form a local frame.

### Proposition 10.60 (p.286)

**Proposition 10.60.** For manifolds $\mathcal{M}$ and $\mathcal{N}$ of dimensions $m$ and $n$, the disjoint union of linear maps from the tangent spaces of $\mathcal{M}$ to those of $\mathcal{N}$,

$$\mathcal{L}(\mathrm{T}\mathcal{M}, \mathrm{T}\mathcal{N}) = \{(x, y, \mathcal{L}) \mid x \in \mathcal{M},\ y \in \mathcal{N} \text{ and } \mathcal{L} \colon \mathrm{T}_x\mathcal{M} \to \mathrm{T}_y\mathcal{N} \text{ is linear}\},$$

is itself a manifold, with charts as follows. Given charts $(\mathcal{U}, \varphi)$, $(\mathcal{V}, \psi)$ and their coordinate frames,

$$\Phi(x, y, \mathcal{L}) = (\varphi(x), \psi(y), \mathrm{Mat}(\mathcal{L})) \in \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R}^{n \times m}$$

is a chart on $\pi^{-1}(\mathcal{U} \times \mathcal{V})$, where $\mathrm{Mat}(\mathcal{L})$ is the matrix in these frames and $\pi(x, y, \mathcal{L}) = (x, y)$.

### Definition 10.61 (p.286)

**Definition 10.61.** Given a manifold $\mathcal{M}$, let $\mathcal{V}$ be open in $\mathcal{M} \times \mathcal{M}$ such that $(x, x) \in \mathcal{V}$ for all $x \in \mathcal{M}$. A transporter on $\mathcal{V}$ is a smooth map

$$\mathrm{T} \colon \mathcal{V} \to \mathcal{L}(\mathrm{T}\mathcal{M}, \mathrm{T}\mathcal{M}) \colon (x, y) \mapsto \mathrm{T}_{y \leftarrow x}$$

such that $\mathrm{T}_{y \leftarrow x}$ is linear from $\mathrm{T}_x\mathcal{M}$ to $\mathrm{T}_y\mathcal{M}$ and $\mathrm{T}_{x \leftarrow x}$ is the identity.

### Propositions 10.62–10.63 (pp.286–287)

**Proposition 10.62.** For a transporter T on $\mathcal{V}$, let $\mathcal{V}'$ be the set of pairs $(x, y) \in \mathcal{V}$ such that $\mathrm{T}_{x \leftarrow y}$ is invertible. Then, the maps $\mathrm{T}'_{y \leftarrow x} = (\mathrm{T}_{x \leftarrow y})^{-1} \colon \mathrm{T}_x\mathcal{M} \to \mathrm{T}_y\mathcal{M}$ define a transporter $\mathrm{T}'$ on $\mathcal{V}'$.

**Proposition 10.63.** Let T be a transporter on a Riemannian manifold $\mathcal{M}$, with domain $\mathcal{V}$. Then, $\mathrm{T}'$ defined by the maps $\mathrm{T}'_{y \leftarrow x} = (\mathrm{T}_{x \leftarrow y})^* \colon \mathrm{T}_x\mathcal{M} \to \mathrm{T}_y\mathcal{M}$ is a transporter on $\mathcal{V}$. Here $*$ denotes the metric adjoint.

### Proposition 10.64 (pp.287–288)

**Proposition 10.64.** For a retraction R on a manifold $\mathcal{M}$, let $\mathcal{T}$ be a neighborhood of the zero section of $\mathrm{T}\mathcal{M}$ such that $E(x, v) = (x, \mathrm{R}_x(v))$ is a diffeomorphism from $\mathcal{T}$ to $\mathcal{V} = E(\mathcal{T})$—such neighborhoods exist by Corollary 10.27. Its inverse is $(x, y) \mapsto (x, \mathrm{R}_x^{-1}(y))$.

1. Assume $\mathcal{T}_x = \{v \in \mathrm{T}_x\mathcal{M} : (x, v) \in \mathcal{T}\}$ is star-shaped around the origin for all $x$. Parallel transport along retraction curves defines a transporter on $\mathcal{V}$ via $\mathrm{T}_{y \leftarrow x} = \mathrm{PT}^c_{1 \leftarrow 0}$, where $c(t) = \mathrm{R}_x(tv)$ and $v = \mathrm{R}_x^{-1}(y)$.
2. The differentials of the retraction define a transporter on $\mathcal{V}$ via $\mathrm{T}_{y \leftarrow x} = \mathrm{DR}_x(v)$, where $v = \mathrm{R}_x^{-1}(y)$.

### Proposition 10.66 (p.288)

**Proposition 10.66.** For an embedded submanifold $\mathcal{M}$ of a Euclidean space $\mathcal{E}$, the maps

$$\mathrm{T}_{y \leftarrow x} = \mathrm{Proj}_y\big|_{\mathrm{T}_x\mathcal{M}},$$

define a transporter on $\mathcal{M} \times \mathcal{M}$, where $\mathrm{Proj}_y \colon \mathcal{E} \to \mathrm{T}_y\mathcal{M}$ is the orthogonal projector.

### Definition 10.68 (pp.290–291)

**Definition 10.68.** A vector transport is a smooth map $(x, u, v) \mapsto \mathrm{VT}_{(x,u)}(v)$ from

$$\mathrm{T}\mathcal{M} \oplus \mathrm{T}\mathcal{M} = \{(x, u, v) : x \in \mathcal{M} \text{ and } u, v \in \mathrm{T}_x\mathcal{M}\}$$

to $\mathrm{T}\mathcal{M}$, satisfying the following for some retraction R on $\mathcal{M}$:

1. $\mathrm{VT}_{(x,u)}$ is a linear map from the tangent space at $x$ to the tangent space at $\mathrm{R}_x(u)$ for all $(x, u) \in \mathrm{T}\mathcal{M}$; and
2. $\mathrm{VT}_{(x,0)}$ is the identity on $\mathrm{T}_x\mathcal{M}$ for all $x \in \mathcal{M}$.

Equivalently, $\mathrm{VT} \colon \mathrm{T}\mathcal{M} \to \mathcal{L}(\mathrm{T}\mathcal{M}, \mathrm{T}\mathcal{M})$ is smooth with the same linearity and identity conditions. A transporter and a retraction yield $\mathrm{VT}_{(x,u)} = \mathrm{T}_{\mathrm{R}_x(u) \leftarrow x}$. The converse need not hold: $\mathrm{VT}_{(x,u)} \neq \mathrm{VT}_{(x,w)}$ may occur even when $\mathrm{R}_x(u) = \mathrm{R}_x(w)$. A vector transport does yield a transporter locally where $(x, u) \mapsto (x, \mathrm{R}_x(u))$ has a smooth inverse (Corollary 10.27).